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Machine learning-based constitutive model for J2- plasticity

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ABSTRACT

This research aims to propose a machine learning (ML)-based constitutive model to predict elastoplastic behavior for J2-plasticity. An artificial neural network (ANN) was constructed to replace the nonlinear stress-integration scheme conducted in the conventional theoretical constitutive model under isotropic hardening and associated flow rule. The training dataset required for the ANN Model was numerically generated based on the conventional return mapping scheme in the principal stress space. The training has been effectively carried out with one element simulation along all the possible plastic loading paths for problem independent training. A conventional theoretical method is used for the unloading procedure. Therefore, ANN is selectively utilized only for nonlinear plastic loading while keeping linear elastic loading and the unloading with a physics-based model. After one element training, the ML-based constitutive model was implemented in Abaqus User MATerial (UMAT) and its performance was verified. For this purpose, one element and tensile test simulations were applied to examine the accuracy of the ANN-based model. Also, for fully nonlinear strain-paths, a circular cup drawing simulation was applied to predict the cup profiles which was compared with that to the conventional J2 plasticity. It was concluded that the simulation results predicted from the ANN-based model show good agreement with those from the conventional J2-based constitutive model. Also, according to simulation time, the ANN-based model shows an advantage in computational efficiency.

1. Introduction

Machine learning is one of the statistical modeling techniques that make a computer system learn or recognize the implicit patterns of the given data without any explicit description. Using the data-driven nature of machine learning, abstract information or theoretically unknown phenomena can be modeled. For this reason, machine learning emerges as an important technology in various engineering fields, such as computer vision, automatic driving, robotics, etc. An artificial neural network (ANN) is an algorithm of the machine learning group, which emulates the human brain. Among diverse machine learning algorithms, ANN is overwhelmingly being popular due to its superior modeling performance and a wide range of applications as a universal approximator (Hornik et al., 1989). McCulloch et al. (1943) proposed the basic mechanism to implement the role of an actual biological neuron by a node inside ANN (see Rumelhart et al., 1986) which built the foundation of optimization for ANN with Backpropagation of error.

Recently, the application of ANN for the field of manufacturing has gradually increased. Because of the data-driven nature of ANN models, researchers can describe a functional relationship among the process variables without any physical background. Viswanathan et al. (2003) applied a neural network along with a two-step trajectory of the binder force to control spring back during a channel

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forming process. Zhao et al. (2005) utilized a neural network to identify material properties and friction coefficient for a deep drawing with an axisymmetric workpiece. Chokshi et al. (2017) developed an ANN-based model predicting the phase distribution during tailored hot stamping of 22MnB5 boron steel. Christoph et al. (2020) described the constitutive behavior of anisotropic elastic-plastic foam structures using a hybrid multi-scale neural network approach. Similarly, Usman et al. (2019) proposed the ANN framework to model stress-strain and texture evolution in single and polycrystal AA6063-T6 under tension and shear. The applicability of machine learning for predicting stress hotspots in different kinds of FCC and HCP materials was demonstrated by Ankita et al. (2018, 2019).

Particularly in the metal plasticity field, ANN has widely been utilized for material modeling. Many researchers describe the hardening behavior of metals with ANN taking the effects of strain rates and temperatures into account. Li et al. (2012) predicted the flow behavior of low alloy steel at elevated temperatures and different strain rates with ANN, and the results from ANN showed a good agreement with experimental data. Lin et al. (2008) used ANN to predict the flow behavior of medium carbon and low alloy steel during hot deformation. Li et al. (2019) used ANN to predict the flow stress of DP800 steel at various temperatures and strain rates and conducted a dog-bone tensile test simulation with the ANN-based hardening model. In these studies, ANN-based hardening models predict hardening behavior at various temperatures and strain rates that could not be described as conventional rate- and temperature-dependent hardening models such as the Johnson-Cook model (see Johnson et al., 1983). Ali et al. (2019) used ANN to predict flow behavior and pole figure of polycrystalline metals under uniaxial tension and simple shear, and the results from ANN matched up with the results from the crystal plasticity finite element (FE) method.

Unlike ANN-based hardening models, ANN-based constitutive models were studied by a few researchers (Ghaboussi et al., 1991; Palau, 2012). Ghaboussi et al. (1991) published a pioneering paper, where they suggested an ANN-based constitutive model for plane concrete under biaxial monotonic loading and uniaxial cyclic loading. The ANN model proposed by Ghaboussi et al. (1991) predicts several loading paths successfully under the biaxial loading condition. However, because they used only a few experimental data for training, the training data do not represent the overall material behavior of plane concrete. Therefore, their ANN model could not predict untrained loading paths. Later, Palau et al. (2012) developed the ANN-based constitutive model to describe elastoplastic behavior in the plane strain condition. Besides, they implemented their ANN model into a commercial FE code (ABAQUS/Explicit). They predicted various types of nonlinear loading paths in single element simulation with the ANN model. However, similarly to the work done by Ghaboussi et al. (1991), their ANN model could not predict untrained loading paths accurately. Consequently, they did not report the multi-element simulation result with their ANN model. To apply an ANN-based constitutive model for practical FE simulation, the ANN-based constitutive model needs to ensure accuracy for any arbitrary loading paths, even for untrained loading paths.

In this paper, an ANN-based constitutive model, which is possible to predict any arbitrary loading path accurately, was developed. The developed model replaces the nonlinear plastic correction procedure for the plastic loading path during the stress integration method while keeping the elastic linear and loading and unloading with a conventional physics-based scheme. That way the training for all the plastic loading paths can be efficiently achieved with one element training. Therefore, ANN can contribute to the most time-consuming iterative procedure during stress integration by using ANN selectively by keeping the advantages of a physics-based model. The developed model is problem independent. For FE analysis, constitutive equations are solved in the time-discrete domain. To reduce the required size of training data, stress integration is performed in the principal stress domain. The solution algorithm with the proposed ANN-based constitutive model will be proposed in Section 2. Training data were numerically generated by the conventional theoretical return mapping model. Overfitting issue was observed during the training process, thus a suitable training dataset was identified to prevent overfitting in Section 3.1. Furthermore, a case study on neural architectures was performed to investigate an efficient neural architecture in Section 3.2. After training had been completed, the ANN model was implemented with UMAT (User MATerial) interface for commercial software, ABAQUS2016/STANDARD. For verification, single-element simulations under three typical loading paths: uniaxial tension, simple shear, and plane strain and dog-bone tensile test simulation were conducted. Furthermore, in order to show the validity for any nonlinear strain paths, the proposed ANN-based constitutive model was applied to a cup drawing simulation. The cup height profile and CPU time were compared to those for the conventional theoretical constitutive model. The aforementioned simulation results will be discussed in Section 4. Finally, conclusions and future work will be summarized in Section 5.

2. An artificial neural network-based constitutive model

The return mapping scheme is a popular solution algorithm to solve the governing equation of associated plastic evolution while satisfying the consistency condition. In this paper, the proposed artificial neural network (ANN)-based constitutive model replaces the aforementioned theoretical constitutive equations. An ANN achieves modeling performance by using the recognized patterns among numerous data. In other words, an ANN can generally describe the unknown relationship among variables through the statistical analysis of the given data themselves without any physics. However, in elastic prediction and plastic connection procedure, only the plastic connection scheme which involves nonlinear iterations is replaced with the ANN model which makes the developed model to help the physics-based model in terms of robustness and speed in this study.

2.1. *Internal mechanism of artificial neural network*

Artificial neural network (ANN) is mainly constructed by three types of layers: (1) input layer, (2) output layer, and (3) hidden layers between the input and output layers. Each layer contains a number of nodes and the node connectivity is adjusted by weight and bias parameters. In the operating process, each node receives the information from the preceding layer with two steps: (1) receiving information and (2) activation. First, the information is delivered in the form of the weighted sum: the linear combination of nodal values in the preceding layer by the weight and bias parameters. In this paper, He et al. (2015) initialization with the normal distribution is utilized for weight initialization. After the delivery of the information, the current node is activated by Rectified Linear Unit (ReLU) (see Nair et al., 2014) function called an activation function.

Assuming an ANN describes the nonlinear mapping from input variables $N^{(i)}$ to output variables $N^{(o)}$, ANN computes along with feed-forward directions from the input layer to the output layer as the following step:

1. Input variables are normalized to avoid ill-conditioning for optimization. Then, the input layer vector becomes

$$\boldsymbol{I} = [x_1, x_2, \cdots, x_{N^{(i)}}]^T \tag{1}$$

2. The first hidden layer takes the weighted sum between the input layer and the first hidden layer and is activated by the ReLU function, f.

$$H_i^{(1)} = f\left(\sum_{j=1}^{N^{(i)}} W_{ij}^{(1)} x_j + b_i^{(1)}\right) \textit{for } i = 1, 2, \cdots, N^{(1)} \tag{2}$$

where $H_i^{(1)}$,$W_{ij}^{(1)}$ and, $b_i^{(1)}$ represent the activated nodal values of the first hidden layer, weight parameters, and bias parameters between the first hidden and input layers, respectively. Superscripts indicate the order of the hidden layers, particularly superscript '*1*' indicates the first hidden layer, subscript '*i*' and '*j*' indicate the order of the components of the activated hidden and preceding hidden

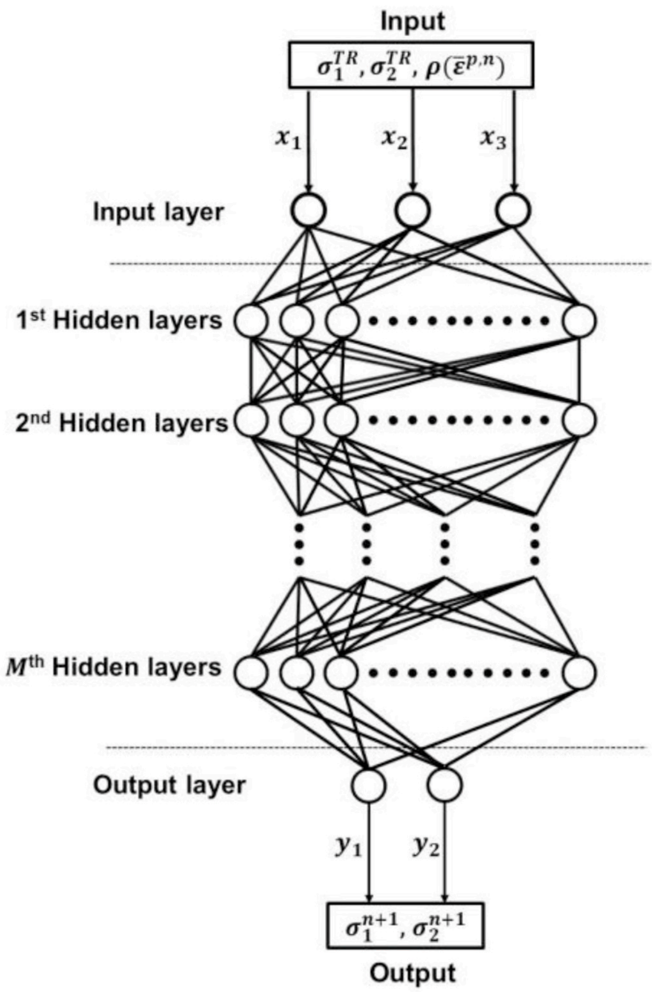

Fig. 1. The neural architecture of the proposed ANN-based constitutive model.

layers, respectively. $N^{(1)}$ is the number of hidden nodes in the first hidden layer.

3. In succession, activation of subsequent $M-1$ hidden layers propagate toward the output layer.

$$\begin{aligned}
&\textit{activation of } 2^{nd} \textit{ hidden layer}: \ H_i^{(2)} = Relu\sum_{j=1}^{N^{(1)}} W_{ij}^{(2)} H_j^{(1)} + b_i^{(2)} \ for\ i = 1, 2, \cdots, N^{(2)} \\
&\textit{activation of } 3^{rd} \textit{ hidden layer}: \ H_i^{(3)} = Relu\sum_{j=1}^{N^{(2)}} W_{ij}^{(3)} H_j^{(2)} + b_i^{(3)} \ for\ i = 1, 2, \cdots, N^{(3)} \\
&\vdots \\
&\textit{activation of } M^{th} \textit{ hidden layer}: \ H_i^{(M)} = Relu\sum_{j=1}^{N^{(M-1)}} W_{ij}^{(M)} H_j^{(M-1)} + b_i^{(M)} \ for\ i = 1, 2, \cdots, N^{(M)}
\end{aligned} \tag{3}$$

where M is the total number of the hidden layers.

4. Finally, the output variables are computed from the M^{th} hidden layer as the following equation,

$$y_i = \sum_{j=1}^{N^{(M)}} W_{ij}^{(o)} H_j^{(M)} + b_i^{(o)} \tag{4}$$

As shown in Eq. (4), the output variables are computed by the linear combination of the activated nodal values in the last hidden layer. Note that the activation function is not utilized for the output layer. Also, the Adaptive Moment Estimation (ADAM) (see Kingma et al., 2014) optimizer is utilized to reduce the cost function in our particular ANN model. Adam optimization is a stochastic gradient descent method that is based on an adaptive estimation of first-order and second-order moments. The mean square error (MSE) was employed as the cost function. The ANN model is trained, such that the MSE is minimized over several epochs or iterations. If no improvement is observed in the value of cost function over 50 consecutive epochs the training is stopped.

2.2. *Solution algorithm of the artificial neural network-based constitutive model*

The proposed ANN-based constitutive model describes the nonlinear functional relationship among two trial principal stresses, present flow stresses, and two updated principal stresses in the time-discrete domain as shown in Fig. 1. The proposed ANN-based constitutive model replaces the plastic corrector step in the return mapping algorithm for plane stress plasticity (see Simo et al., 1998).

In the conventional approach, return mapping is directly performed in the projected plane-stress subspace, such that a generic plane stress tensor with the corresponding plane stress subspace $\mathbb{S}_p$ is defined as:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{yy} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \ \mathbb{S}_p = \left\{ \boldsymbol{\sigma} \middle| \sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0 \right\} \tag{5}$$

Considering the solution algorithm for the time-discrete constitutive equations, it is identical to the elastic prediction step in the conventional theoretical approach and the proposed ANN-based constitutive model. Subsequently, the consistency condition is checked to identify whether the given loading path causes plastic evolution with the following equation:

$$R = \overline{\sigma}\left(\boldsymbol{\sigma}^{TR}\right) - \rho\left(\overline{\varepsilon}^{p,n}\right) \tag{6}$$

where R is the residual for the consistency condition, $\boldsymbol{\sigma}^{TR}$ is the trial stress tensor, $\overline{\varepsilon}^{p,n}$ is the present effective plastic strain, ρ is the corresponding flow stress and $\overline{\sigma}$ is the equivalent stress. When the computed value of Eq. (6) is negative or zero, the updated stress is the same as the trial stress. Then, the plastic strain is not accumulated, and the plastic corrector step is skipped. In contrast, if it is larger than zero, the plastic corrector step requires the plastic correction by the return mapping method in the constrained plane-stress subspace.

In the conventional approach, an iterative scheme such as the Newton-Raphson algorithm may be employed to compute the plastic multiplier$\Delta\gamma$to obtain updated plane stress tensor, such that:

$$\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{TR} - \mathbf{C}^e \Delta\gamma \mathbf{P} \boldsymbol{\sigma}^{n+1} \tag{7}$$

where $\boldsymbol{\sigma}^{n+1}$ is the updated stress tensor, $\boldsymbol{\sigma}^{TR}$ is the trial stress tensor,$\boldsymbol{C}^e$ is the plane stress elasticity matrix and $\mathbf{P}$ is the matrix defined by,

$$\mathbf{P} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix} \tag{8}$$

On the other hand, with $R > 0$, the proposed ANN-based constitutive model operates with the following steps:

1. Compute spectral decomposition of the trial stress, i.e.,

$$\boldsymbol{\sigma}^{TR} = \sum_{A=1}^{2} \sigma_A^{TR} \boldsymbol{n}^A \otimes \boldsymbol{n}^A \tag{9}$$

2. Compute the updated principal stress with the ANN-based constitutive model, i.e.,

$$ANN : \left\{ \begin{array}{c} \sigma_1^{TR} \\ \sigma_2^{TR} \\ \rho(\overline{\varepsilon}^{p,n}) \end{array} \right\} \rightarrow \left\{ \begin{array}{c} \sigma_1^{n+1} \\ \sigma_2^{n+1} \end{array} \right\} \tag{10}$$

3. Update the effective plastic strain assuming the consistency is maintained, i.e.,

$$\rho(\overline{\varepsilon}^{p,n+1}) = \overline{\sigma}\left(\boldsymbol{\sigma}^{n+1}\right) \tag{11}$$

The effective plastic strain is computed as,

$$\overline{\varepsilon}^{p,n+1} = \left(\frac{\rho}{K}\right)^{\frac{1}{n}} - \varepsilon_0 \tag{12}$$

where K, ε_0, and $\boldsymbol{n}$ are constants of Swift law computed based on materials flow stress curve.

4. Perform the rotation transformation of the updated stress tensor to the material coordinate, i.e.,

$$\boldsymbol{\sigma}^{n+1} = \sum_{A=1}^{2} \sigma_A^{n+1} \boldsymbol{n}^A \otimes \boldsymbol{n}^A = \sum_{i=1}^{2} \sum_{j=1}^{2} \sigma_{ij}^{n+1} \boldsymbol{e}^i \otimes \boldsymbol{e}^j \tag{13}$$

3. Training data and neural architecture with one element FEM

For successful training, the suitable training dataset and neural architecture were identified. The most important factor determining the modeling performance of ANN is the quality and quantity of training data. Training data should represent the overall features of the modeling target, otherwise, the overfitting problem occurs during the training process. Before describing the ANN network, a brief overview of the plane stress return mapping scheme was presented in section 3.1. In Section 3.2, the suitable training dataset that prevents the overfitting problem was investigated.

The modeling performance of an ANN is also dependent on a neural architecture. In the neural network, there are several hidden layers, and each layer has a number of nodes. A case study on neural architectures was performed and the best architecture in terms of the modeling accuracy was identified in Section 3.3. Thereafter the training of the ANN model was discussed in section 3.4.

3.1. Review of plane stress return mapping method

Several procedures to find the numerical solution to the plane stress elastoplastic problem was discussed by Neto et al. (2008). The UMAT employed in this work utilizes an implicit return mapping scheme (Simo and Taylor, 1986; Simo, 1998), where the return mapping is performed in the constraint plane-stress subspace. Further, Yoon et al. (2004) presented a plane stress return mapping method for a non-quadratic anisotropic yield function with three stress components (Barlat et al., 2003) where finite element implementation was performed based on the Euler-Backward method with ABAQUS UMAT. This section provides a brief overview of the plane stress algorithm by Simo and Taylor (1986) with the ANN algorithm interface.

$$\begin{aligned} &\dot{\boldsymbol{\varepsilon}} = \dot{\boldsymbol{\varepsilon}}^e + \dot{\boldsymbol{\varepsilon}}^p \\ &\boldsymbol{\sigma} = \mathbf{C}^e \boldsymbol{\varepsilon}^e \\ &f = \frac{1}{2}\boldsymbol{\sigma}^T \mathbf{P} \boldsymbol{\sigma} - \frac{1}{3}\rho^2(\overline{\varepsilon}^p) \\ &\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \mathbf{P} \boldsymbol{\sigma} \\ &\dot{\overline{\varepsilon}}^p = \dot{\gamma}\sqrt{\frac{2}{3}\boldsymbol{\sigma}^T \mathbf{P} \boldsymbol{\sigma}} \end{aligned} \tag{14}$$

where stress vector is $\boldsymbol{\sigma} := \begin{bmatrix} \sigma_{xx} & \sigma_{yy} & \sigma_{xy} \end{bmatrix}^T$, $\sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$, $\boldsymbol{C}^e$ is the plane stress elasticity matrix. Considering the general framework for elastic predictor/return-mapping scheme, the elastic predictor state is computed as,

$$\begin{aligned}&\boldsymbol{\varepsilon}^{n+1}=\boldsymbol{\varepsilon}^{n}+\Delta\boldsymbol{\varepsilon}\\&\boldsymbol{\sigma}^{n+1}=\mathbf{C}^{e}\left(\boldsymbol{\varepsilon}^{n+1}-\boldsymbol{\varepsilon}^{p}\right)\\&f^{TR}=\frac{1}{2}\left(\boldsymbol{\sigma}^{TR}\right)^{T}\mathbf{P}\boldsymbol{\sigma}^{TR}-\frac{1}{3}\rho^{2}\left(\overline{\varepsilon}^{p,TR}\right)\end{aligned}\tag{15}$$

If $f^{TR}\leq 0$ the process is elastic else, the plastic return mapping utilizing a backward-Euler difference scheme requires solving the following equations,

$$\begin{aligned}&\boldsymbol{\varepsilon}^{p,n+1}=\boldsymbol{\varepsilon}^{p,n}+\Delta\gamma\mathbf{P}\boldsymbol{\sigma}^{n+1}\\&\overline{\varepsilon}^{p,n+1}=\overline{\varepsilon}^{p,n}+\Delta\gamma\sqrt{\frac{2}{3}\left(\boldsymbol{\sigma}^{n+1}\right)^{T}\mathbf{P}\boldsymbol{\sigma}^{n+1}}\\&\frac{1}{2}\left(\boldsymbol{\sigma}^{n+1}\right)^{T}\mathbf{P}\boldsymbol{\sigma}^{n+1}-\frac{1}{3}\rho^{2}\left(\overline{\varepsilon}^{p,n+1}\right)=0\end{aligned}\tag{16}$$

where $\Delta\gamma$ is the plastic Lagrange multiplier. Assuming isotropic linear elasticity, the Cauchy stress vector at the end of increment is,

$$\begin{aligned}&\boldsymbol{\sigma}^{n+1}=\boldsymbol{\sigma}^{n}+\mathbf{C}^{e}\Delta\boldsymbol{\varepsilon}^{e}\\&\quad=\boldsymbol{\sigma}^{n}+\mathbf{C}^{e}\left(\Delta\boldsymbol{\varepsilon}-\Delta\boldsymbol{\varepsilon}^{p}\right)\\&\because\boldsymbol{\sigma}^{TR}=\boldsymbol{\sigma}^{n}+\mathbf{C}^{e}\Delta\boldsymbol{\varepsilon}\\&\boldsymbol{\sigma}^{n+1}=\boldsymbol{\sigma}^{TR}-\mathbf{C}^{e}\Delta\gamma\mathbf{P}\boldsymbol{\sigma}^{n+1}\end{aligned}\tag{17}$$

The $\Delta\gamma$, plastic Lagrange multiplier is determined using the Newton-Raphson iterative scheme enforcing the consistency condition at the end of the time-step such that,

$$f(\Delta\gamma):=\frac{1}{2}\xi^{n+1}\left(\Delta\gamma\right)-\frac{1}{3}\rho^{2}\left(\overline{\varepsilon}^{p,n}+\Delta\gamma\sqrt{\frac{2}{3}\xi^{n+1}\left(\Delta\gamma\right)}\right)=0\tag{18}$$

where $\xi^{n+1}=(\boldsymbol{\sigma}^{n+1})^{T}\mathbf{P}\boldsymbol{\sigma}^{n+1}$ and with the solution of $\Delta\gamma$, the stress, plastic strain, and equivalent strain are updated. After the updated stress caused by the material deformation is obtained, thickness strain is updated as,

$$\varepsilon_{33}=-\frac{\nu}{E}\left(\sigma_{xx}+\sigma_{yy}\right)-\left(\varepsilon_{xx}^{p}+\varepsilon_{yy}^{p}\right).\tag{19}$$

A substantial advancement of the plane stress return mapping method is to completely determine the plastic part of the deformation without an iterative estimation of the thickness strain increment as shown in Eq. (19).

Finally, the elastoplastic tangent modulus is defined as,

$$d\boldsymbol{\sigma}^{n+1}=\mathbf{C}^{ep}d\boldsymbol{\varepsilon}^{n+1}\tag{20}$$

where

$$\mathbf{C}^{ep}=\overline{\mathbf{C}}-\frac{\overline{\mathbf{C}}\mathbf{m}_{n+1}\otimes\overline{\mathbf{C}}\mathbf{m}_{n+1}}{\mathbf{m}_{n+1}\overline{\mathbf{C}}\mathbf{m}_{n+1}+h'}$$

$$\overline{\mathbf{C}}=\left[\mathbf{C}^{e^{-1}}+\gamma\frac{\partial\mathbf{m}_{(n+1)}}{\partial\boldsymbol{\sigma}_{(n+1)}}\right]^{-1},\ \mathbf{m}=\frac{\partial\overline{\sigma}}{\partial\boldsymbol{\sigma}},\ \mathbf{C}^{e}=\frac{E}{1-\nu^{2}}\begin{bmatrix}1&\nu&0\\\nu&1&0\\0&0&\frac{1-\nu}{2}\end{bmatrix}$$

On the other hand, the UMAT based on the ANN model focuses on replacing the iterative procedure for finding the updated stress vector. Equations (16)–(18) are replaced by the ANN network as described in Section 2.2. Given the trial stress vector $\boldsymbol{\sigma}^{TR}:=\begin{bmatrix}\sigma_{xx}^{TR}&\sigma_{yy}^{TR}&\sigma_{xy}^{TR}\end{bmatrix}^{T}$, the principal stresses are computed as,

$$\sigma_{1,2}^{TR}=\frac{1}{2}\left(\sigma_{xx}^{TR}+\sigma_{yy}^{TR}\right)\pm\sqrt{\left(\frac{\sigma_{xx}^{TR}-\sigma_{yy}^{TR}}{2}\right)^{2}+\sigma_{xy}^{TR2}}\tag{21}$$

$$\cos 2\theta=\frac{\frac{\sigma_{xx}^{TR}-\sigma_{yy}^{TR}}{2}}{\sqrt{\left(\frac{\sigma_{xx}^{TR}-\sigma_{yy}^{TR}}{2}\right)^{2}+\sigma_{xy}^{TR2}}},\ \sin 2\theta=\frac{\sigma_{xy}^{TR}}{\sqrt{\left(\frac{\sigma_{xx}^{TR}-\sigma_{yy}^{TR}}{2}\right)^{2}+\sigma_{xy}^{TR2}}}\tag{22}$$

The above trial principal stresses along with flow stress are fed into the ANN network as indicated in equation (10) to calculate the updated principal stresses. Thereafter, the updated principal stresses are transformed back to the material coordinates using the following relations,

$$
\begin{aligned}
\sigma_{xx}^{n+1} &= \frac{1}{2}\left(\sigma_1^{n+1} + \sigma_2^{n+1}\right) + \frac{1}{2}\left(\sigma_1^{n+1} - \sigma_2^{n+1}\right)\cos 2\,\theta \\
\sigma_{yy}^{n+1} &= \frac{1}{2}\left(\sigma_1^{n+1} + \sigma_2^{n+1}\right) - \frac{1}{2}\left(\sigma_1^{n+1} - \sigma_2^{n+1}\right)\cos 2\,\theta \\
\sigma_{xy}^{n+1} &= -\frac{1}{2}\left(\sigma_1^{n+1} - \sigma_2^{n+1}\right)\sin 2\,\theta
\end{aligned}
\tag{23}
$$

3.2. An investigation into a suitable training dataset

Training data were numerically generated based on the conventional theoretical return mapping scheme. Recalling the ANN-based constitutive model describes the nonlinear functional relationship among the three input variables (σ_1^{TR},σ_2^{TR} and $\rho(\overline{\varepsilon}^p)$) and the two output variables (σ_1 and σ_2) in the time-discrete domain, the two trial stresses (σ_1^{TR} and σ_2^{TR}) are computed from the prescribed current flow stress $\rho(\overline{\varepsilon}^p)$ and two updated stresses (σ_1 and σ_2) with following equations:

$$
\begin{bmatrix} \sigma_1^{TR} \\ \sigma_2^{TR} \end{bmatrix} = \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} + \Delta\gamma \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} (2\sigma_1 - \sigma_2)/2\overline{\sigma} \\ (2\sigma_2 - \sigma_1)/2\overline{\sigma} \end{bmatrix} \tag{24}
$$

where $\Delta\gamma$ is the plastic parameter for the given loading path, E is the Elastic modulus and ν is the Poisson ratio. The size of a training dataset is dependent on two design variables of the data generation scheme. The first design variable is the density of data on the hardening curve, which is relative to the input variable $\rho(\overline{\varepsilon}^p)$ shown in Fig. 2. The non-uniform grid on $\overline{\varepsilon}^p$-axis is utilized, and interval among non-uniform grid ($\overline{\varepsilon}_{n+1}^p - \overline{\varepsilon}_n^p$), where the subscript n indicates the order of a data point on the hardening curve, should be determined. The hardening curve using Swift law takes the form,

$$
\rho\left(\overline{\varepsilon}^{p,n+1}\right) = K^*\left(\varepsilon_0 + \overline{\varepsilon}^{p,n+1}\right)^n [MPa] \tag{25}
$$

where K, ε_0, and n are constants computed based on materials flow stress curve.

The second design variable is the density of data on the yield surface at the given hardening, which is related to the two output variables (σ_1 and σ_2) shown in Fig. 3. The two updated principal stresses are represented in the polar coordinate as follows:

$$
\begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \begin{bmatrix} \sigma_\theta \cos\theta \\ \sigma_\theta \sin\theta \end{bmatrix} \quad -\frac{\pi}{4} \le \theta \le \frac{3\pi}{4} \tag{26}
$$

where θ is the angle from the σ_1-axis. Stress is equally discretized with the constant difference in the angle $\Delta\theta$.

The size of the dataset and the complexity of the ANN network are the primary design factors as it poses a tradeoff. If the training dataset cannot represent the overall features of the modeling target, in case of insufficient data, a deep ANN model may overfit the modeling target. Contrary, If the training dataset is too big, a shallow ANN model may overfit the modeling target. The adjective "deep" in deep learning usually indicates networks with multiple layers. To prevent the overfitting or underfitting problem, a sufficient amount of training data is required. A deep neural architecture with 2 layers/30 nodes was employed for the selection of the size of the training set. Further discussion of the neural architecture can be found in section 3.2. To investigate the suitable density of training data on the hardening curve, three distinct datasets were compared as shown in Table 1.

For the first dataset, the interval among the non-uniform grid is a random value in the range of [10^{-3}, 10^{-2}] and the number of data points on the hardening curve is 20. Hence, the effective plastic strain starting from the value of 0 in the hardening curve is discretized into 20 points up to the maximum value of 0.1. From the first set to the third set, the interval among the non-uniform grid decreases 10

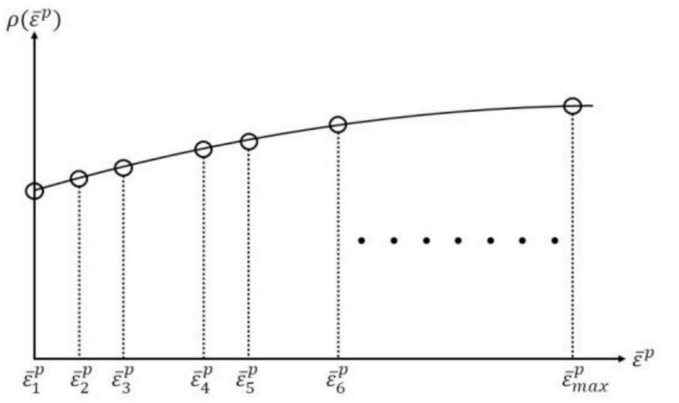

Fig. 2. Schematic representation of the density of training data on the hardening curve.

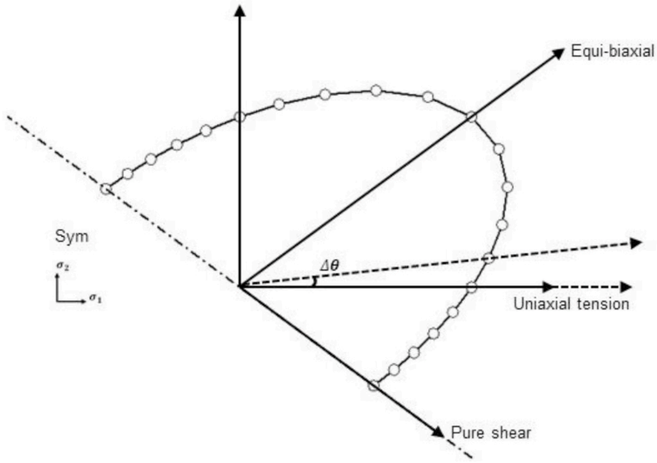

Fig. 3. Schematic representation of the number of stress points on updated (half) yield surface.

times, and accordingly, the number of data points on the hardening curve increases 10 times. After 10 epochs of training, the maximum error of the mean of 2 output variables is plotted with respect to the epoch in Fig. 4.

When the ANN model is trained by the first dataset, the maximum error for the ANN output σ_1 decreases as training continues. However, the maximum error for the second and third dataset increase, which is the observation of the overfitting. Additionally, the maximum error for all datasets decreases as training continues. Consequently, a suitable interval of the non-uniform grid on $\bar{\varepsilon}^p$-axis is $[10^{-4}, 10^{-3}]$ following that of the second dataset.

Similarly, the suitable density of training data on the updated yield surface is investigated by comparing three datasets as shown in Table 2.

For the first set, the number of stress points on (half) yield surface is 45, and relevant $\Delta\theta = 2.5^{\circ}$.

From the first to the third sets, the number of stress points increases, and accordingly relevant $\Delta\theta$ decreases. After 200 epochs of training, the maximum error of the mean of 2 output variables is plotted with respect to the epoch in Fig. 5.

When the ANN model is trained by the first dataset, the overfitting problem is observed after 180 epochs. When the ANN model is trained by the second datasets, the maximum error for all datasets decreases as training continues. Consequently, a suitable number of stress points on updated (half) yield surface is 180, and relevant $\Delta\theta = 1^{\circ}$. To consider the large deformation regime, the maximum effective plastic strain in training data $\bar{\varepsilon}^p_{max}$ is defined 0.6 and determined the design variables for the training dataset are listed in Table 3. Note that excessively large loading paths are excluded, i.e.,

$$\bar{\sigma}\left(\boldsymbol{\sigma}^{TR}\right) - \rho(\bar{\varepsilon}^p) > 1.2^*(initialyield\ \ stress) \tag{27}$$

3.3. Case study on neural architectures

The modeling performance of a neural network depends on two variables, depth, and wideness of the neural network. The quantitative indicators of depth and wideness are the number of hidden layers and the number of hidden nodes per hidden layer, respectively. For the depth, two levels were compared: two-layer and five-layer architectures. For the wideness, three levels were compared: ten-node, thirty-node- and fifty-node architectures. Accordingly, six distinct neural architectures were compared. The maximum error of six different architectures right after 100 epochs are compared in Fig. 6. For the case of 10 node-architectures, the error is large regardless of depth. If the number of nodes per layer is larger than 30, the degree of error is similar for any architecture. The combination of 5 layers and 30 nodes shows the minimum error among 6 architectures. Comparing the architecture with 5 layers/30 nodes and 5 layers/50 nodes, the error is larger even though the number of nodes increases. Consequently, the architecture with 5 layers and 30 nodes was employed in this paper. Glorot et al. (2011) indicated that linear rectifiers such as Rectified Linear Unit (ReLU) (see Nair et al., 2014), there is no gradient vanishing effect due to activation non-linearities of Sigmoid or Tanh units. Hence, except for the output nodes, the ReLU was employed as the activation function.

Table 1
Three distinct datasets with different density of training data on the hardening curve.

Set	Interval between non-uniform grid ($\bar{\varepsilon}^p_{n+1} - \bar{\varepsilon}^p_n$)	The number of data points on the hardening curve	The maximum effective plastic strain ($\bar{\varepsilon}^p_{max}$)	The number of stress points on the updated yield surface	Plastic accumulation for given loading path
Set1	$[10^{-3}, 10^{-2}]$	20	0.1	180	$\Delta\gamma \in [10^{-6}, 10^{-2}]$
Set2	$[10^{-4}, 10^{-3}]$	200			
Set3	$[10^{-5}, 10^{-4}]$	2000			

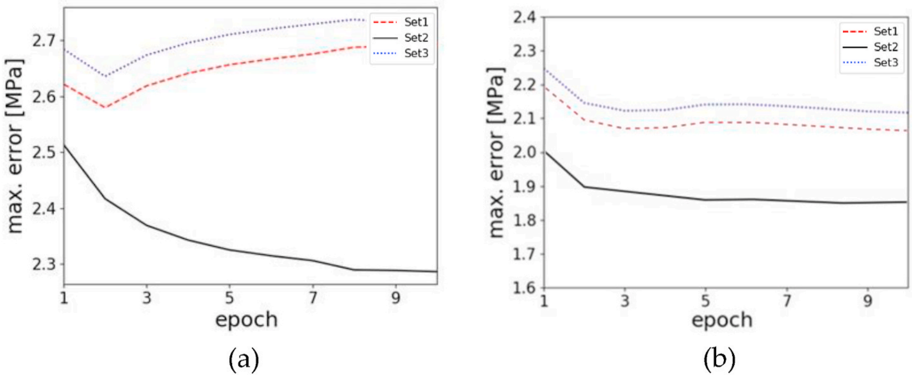

Fig. 4. Maximum mean square error (MSE) of the two output variables (σ_1 and σ_2) versus epoch plot for 3 sets: (a) MSE based on ANN output σ_1; (b) MSE based on ANN output σ_2.

Table 2
Three distinct datasets with different numbers of stress points on the updated yield surface.

Set	Angle between neighboring stress points ($\Delta\theta$)	The number of stress points on the trial yield surface	The maximum effective plastic strain ($\bar{\varepsilon}^p_{max}$)	Interval between non-uniform grid ($\bar{\varepsilon}^p_{n+1} - \bar{\varepsilon}^p_n$)	Plastic accumulation for given loading path
Set1	2.5°	45	0.1	$[10^{-4}, 10^{-3}]$	$\Delta\gamma \in [10^{-6}, 10^{-2}]$
Set2	1°	180			
Set3	0.5°	360			

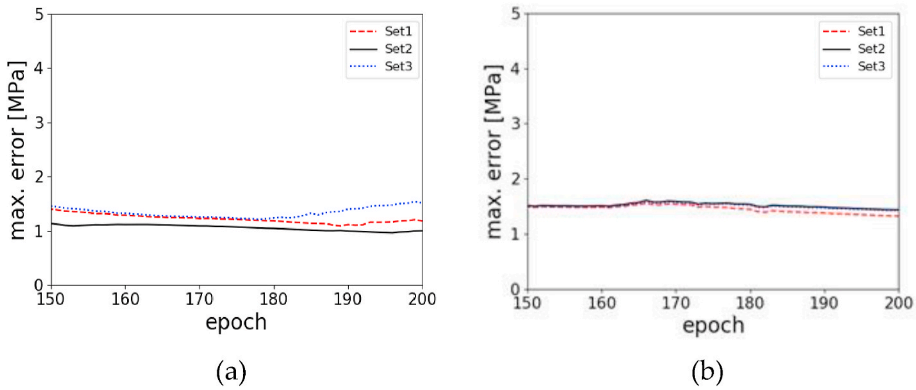

Fig. 5. Maximum mean square error (MSE) of the two output variables (σ_1 and σ_2) versus epoch plot for 3 sets: (a) MSE based on ANN output σ_1; (b) MSE based on ANN output σ_2.

Table 3
Determined design variables for the training dataset.

Density of training data on the hardening curve	$\Delta\bar{\varepsilon}^p \in [10^{-4}, 10^{-3}]$ $\bar{\varepsilon}^p_{max} = 0.6$	1200 points
The number of stress points on the updated yield surface	$\Delta\theta = 1°$	180 points
Plastic accumulation for given loading path	$\Delta\gamma \in [10^{-6}, 10^{-2}]$	–

3.4. Training the ANN model

After the selection of a suitable training dataset and neural architecture, the ANN network was trained and the cost function is monitored for the course of training. The total size of the dataset was 338182, out of which 90% was used for training and 10% was utilized for testing the performance of the ANN model. A fraction of 0.1 of the training set is used as a validation set during the training, hence, avoiding overfitting, by monitoring the cost function of the validation set. Using early stopping criteria, such that if no improvement is observed in the value of cost function for the validation set over 50 consecutive epochs the training is stopped. Thereafter, the training stopped at 412 epochs, and the maximum error of the 2 output variables is plotted with respect to the epoch in Fig. 7.

Since the ANN model is trained, we extracted the weights and biases of the trained model. Thereafter, the ANN-based constitutive model was implemented in Abaqus User MATerial (UMAT) using the mechanism explained in Section 2.1, obtaining Eq. (10). To illustrate the characteristics of the proposed ANN model, one design case has been considered and implemented in MATLAB. A random sample input was chosen from the dataset, such that $\sigma_1^{TR} = 331.75$ MPa, $\sigma_2^{TR} = 167.78$ MPa and $\rho(\bar{\varepsilon}^{p,n}) = 280.84$ *MPa*. The input was fed into the ANN model Blackbox, and the output of each hidden layer was noted as shown in Fig. 8. Inactive neurons are because the ReLU function does not activate for a negative input, and activates for positive input only. The output of the Blackbox was $\sigma_1^{n+1} = 324.26$ *MPa* and $\sigma_2^{n+1} = 165.22$ *MPa*. Contrary to the convolutional neural network in which one can visualize the learned images in-between hidden layers, such observation is rather difficult for numerical data-based ANN models. However, as the training proceeds, the ANN model learns the explicit relationship between its input and output, as observed in section 4.

4. Results and discussion

After training had been completed, the ANN-based constitutive model was implemented in the User MATerial interface (UMAT) for ABAQUS 2016/STANDARD. To verify the accuracy of the proposed ANN-based constitutive model, single-element simulation under four typical loading paths was conducted, and the simulation results were compared with those obtained from the conventional theoretical constitutive model with Euler Backward return mapping scheme. Additionally, to verify the feasibility of the ANN-based constitutive model for multi-element simulation, a dog-bone tensile test simulation was conducted. Finally, the ANN-based constitutive model was applied for a circular cup drawing simulation for fully nonlinear strain paths. The simulation results such as the cup height and simulation time were compared with those obtained from the conventional theoretical constitutive model.

4.1. Verification1: single-element simulation

The ANN model replaces the conventional constitutive model composed of von Mises yield function and Swift isotropic hardening law with the following equation:

$$\rho(\bar{\varepsilon}^p) = 646*(0.025 + \bar{\varepsilon}^p)^{0.227}[MPa] \quad (28)$$

Single-element uniaxial tension (elongated along the x-direction), simple shear, and plane strain (elongated along the x-direction), and cyclic (uniaxial tension-compression-tension) simulation were conducted. The single-element was a four-node element with reduced integration, donated as S4R. The young modulus was 70 GPa with Poisson's ratio of 0.33. The fixed time increment setting was employed with 100 increments. As shown in Fig. 9, the stress responses predicted from the proposed ANN-based constitutive model

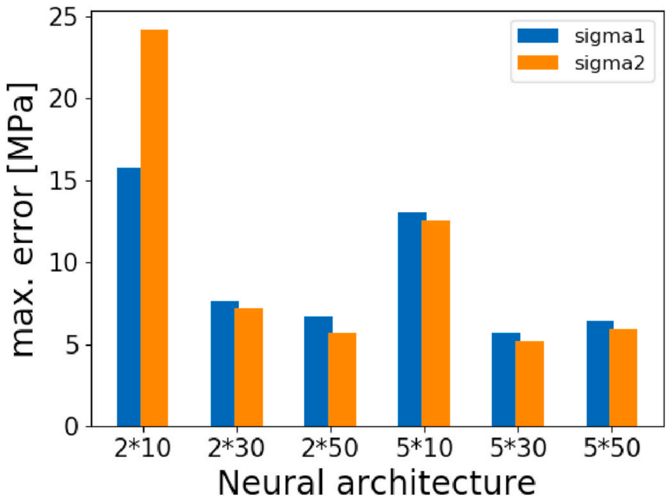

Fig. 6. Comparison of maximum MSE error right after 100 epochs by 6 distinct neural architectures.

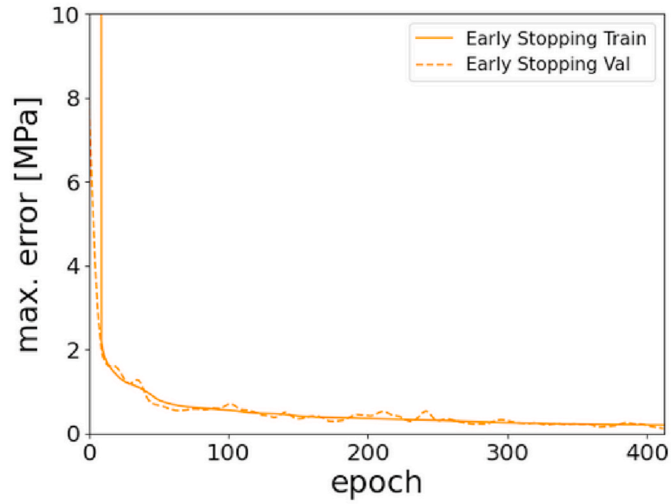

Fig. 7. Maximum error of the 2 output variables using early stopping criteria.

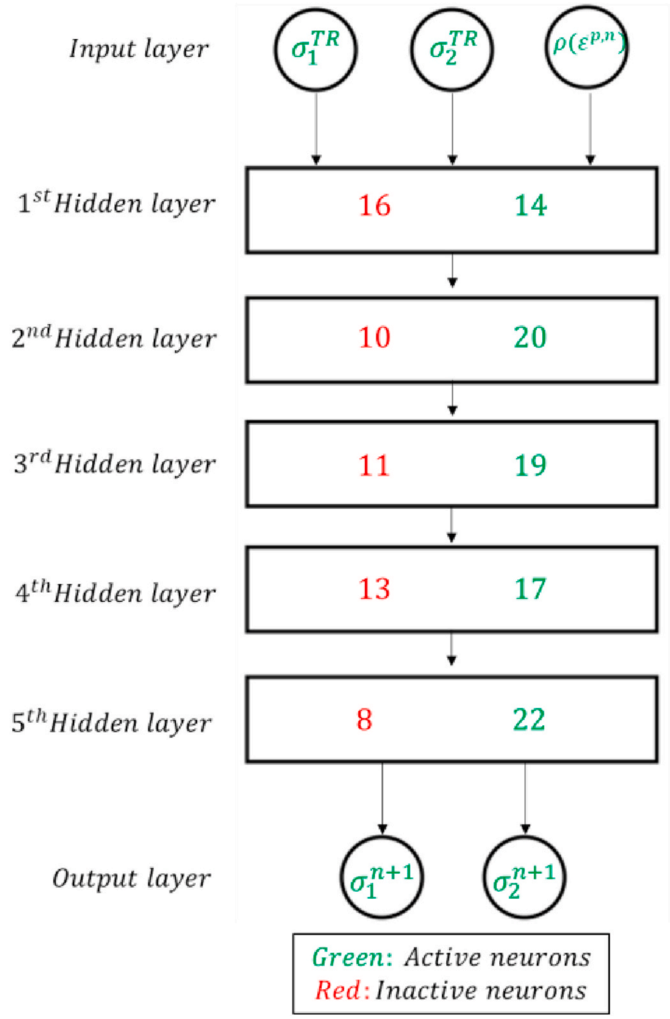

Fig. 8. The Blackbox of the proposed ANN model indicating output of each layer for a random input sample.

show a good agreement with those predicted from the theoretical constitutive model. Comparing with the conventional theoretical constitutive model, the mean relative errors for the four loading paths are less than 1%, and there is no overshooting value. Therefore, the ANN model was successfully able to predict the correct loading path among all the possible loading paths employed in the training set.

4.2. Verification2: dog-bone tensile test simulation

To verify the feasibility of the proposed ANN-based constitutive model on multi-elements simulation, a dog-bone tensile test simulation was conducted. Utilized material properties are the same as those used in Section 4.1. ASTM E8/E8M-16a (2016) sub-size specimen was used for the simulation as shown in Fig. 10. The shell element with reduced integration (S4R) was employed. The total number of elements is 1000. The auto time increment setting was employed with the minimum 0.001 s and maximum 0.02 s for the total time of 1 s. For boundary conditions, the left side of the specimen is fixed, and the right side of the specimen is stretched until 9 mm.

Fig. 11 and Fig. 12 show the comparisons of the effective stress and effective plastic strain predicted from the theoretical and proposed ANN-based constitutive models, respectively. The results show a negligible difference between the proposed ANN-based and theoretical constitutive models. For the dog-bone simulation when compared to the theoretical model the mean relative error of stress responses predicted by the ANN is 0.31%. For effective plastic strain contour, there is no noticeable difference.

For the detailed comparisons, five points were selected from the dog-bone specimen, three in the gauge section, and two in the fillet region as shown in Fig. 13. In the gauge section, the longitudinal stress predicted from the ANN model exactly follows the targeted constitutive equation, obtained from the analytical model as shown in Fig. 14 (a). Fig. 14 (b) shows the stress-triaxiality in the gauge section. The stress-triaxiality predicted by the ANN model maintains the value of $\frac{1}{3}$, which means that the ANN model well describes the uniaxial tension path in the gauge section.

In the fillet regions of a dog-bone specimen, the specimen undergoes more complex nonlinear strain paths, which contain both normal stress and shear stress. Fig. 15 compares two normal stresses and the shear stresses predicted from the theoretical and ANN-

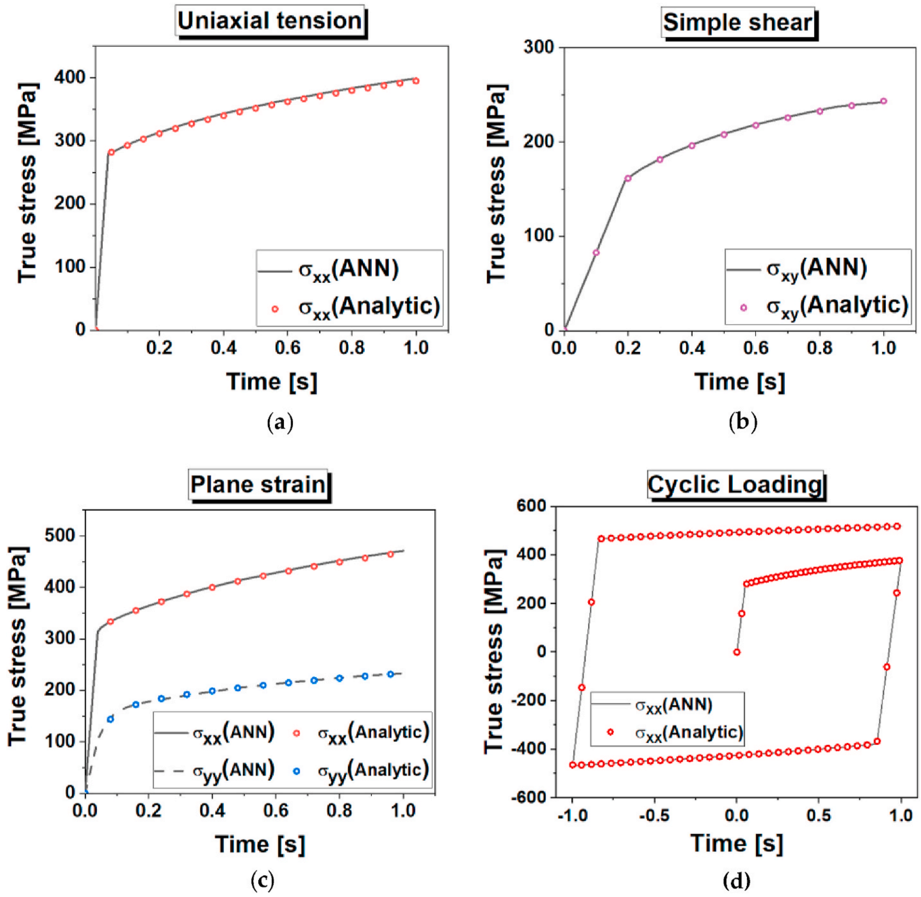

Fig. 9. Comparison of stress responses predicted by the ANN model and the theoretical model for the four typical loading paths: (**a**) uniaxial tension; (**b**) simple shear; (c) plane strain; (d) cyclic loading.

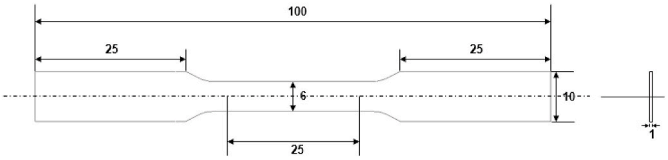

Fig. 10. ASTM E8/E8M-16a standard for the sub-size dog-bone specimen.

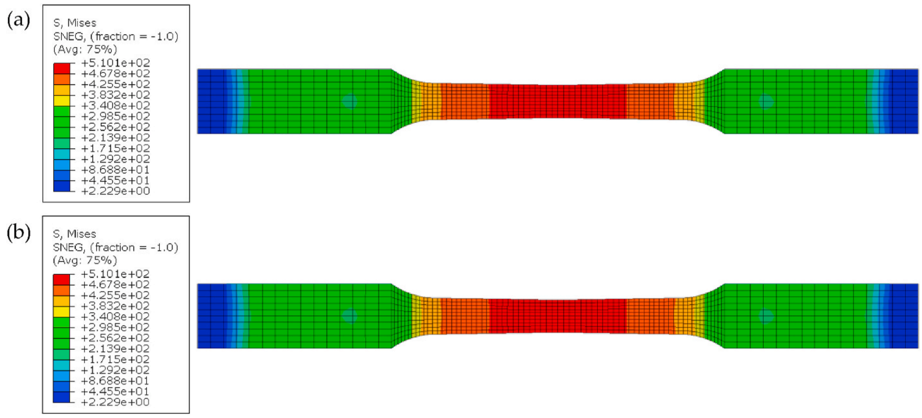

Fig. 11. Comparison of effective stress contours for the dog-bone tensile test simulation (**a**) Effective stress predicted by the theoretical constitutive model; (**b**) Effective stress predicted by the ANN-based constitutive model.

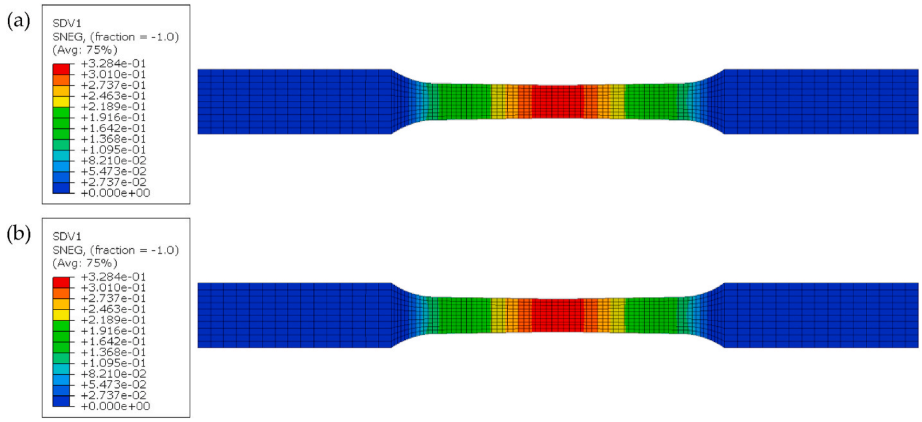

Fig. 12. Comparison of effective plastic strain contours for the dog-bone tensile test simulation: (**a**) Effective plastic strain predicted by the ANN-based constitutive; (**b**) Effective plastic strain predicted by the ANN-based constitutive model.

based constitutive models in the material embedded coordinate and both results are in good agreement.

4.3. Application: circular cup drawing simulation

The proposed ANN-based constitutive model was applied to a circular cup drawing simulation. The detailed dimensions and process variables for circular cup drawing simulation were set by referring to Yoon et al. (2006). The type of elements for the blank sheet is S4R, and 360 elements with 394 nodes are utilized as shown in Fig. 16. The auto time increment setting was employed with the regulation of maximum time increment $5*10^{-3}$ for the total time of 1 s.

Cup height profiles predicted from both the theoretical and ANN models have a good agreement as shown in Fig. 17. Because of the isotropic nature of J2 plasticity, the cup height profile shows a straight line.

The effective stress and effective plastic strain contours were compared in Figs. 18 and 19. In the case of effective stress contours, there is max. 10 MPa difference at the bottom region of the cup. For the cup drawing simulation when compared to the theoretical model the mean relative error of stress responses predicted by the ANN is 0.77%. For effective plastic strain contour, there is no

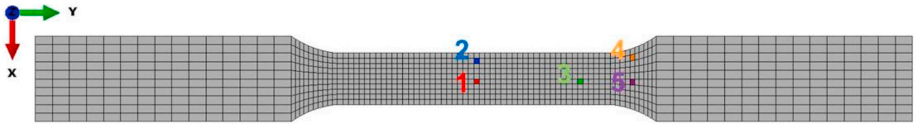

Fig. 13. Five points where the stress responses were extracted.

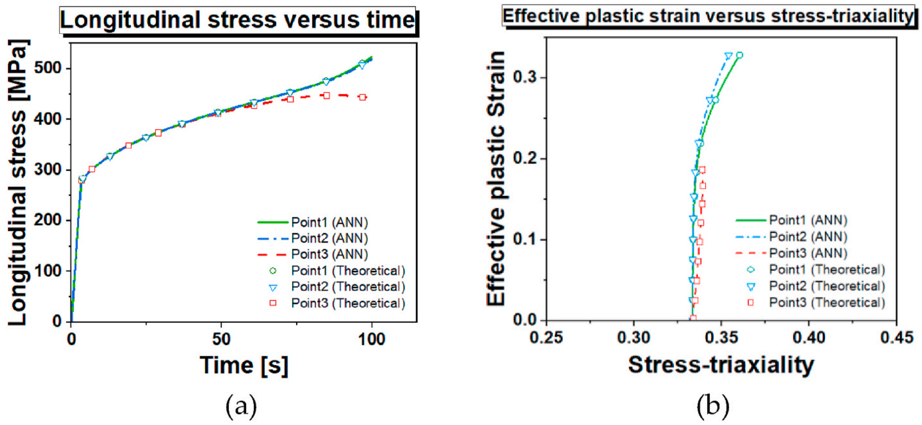

Fig. 14. The results of three points in the gauge section predicted by the ANN-based constitutive model: (**a**) longitudinal stress response; (**b**) stress-triaxiality.

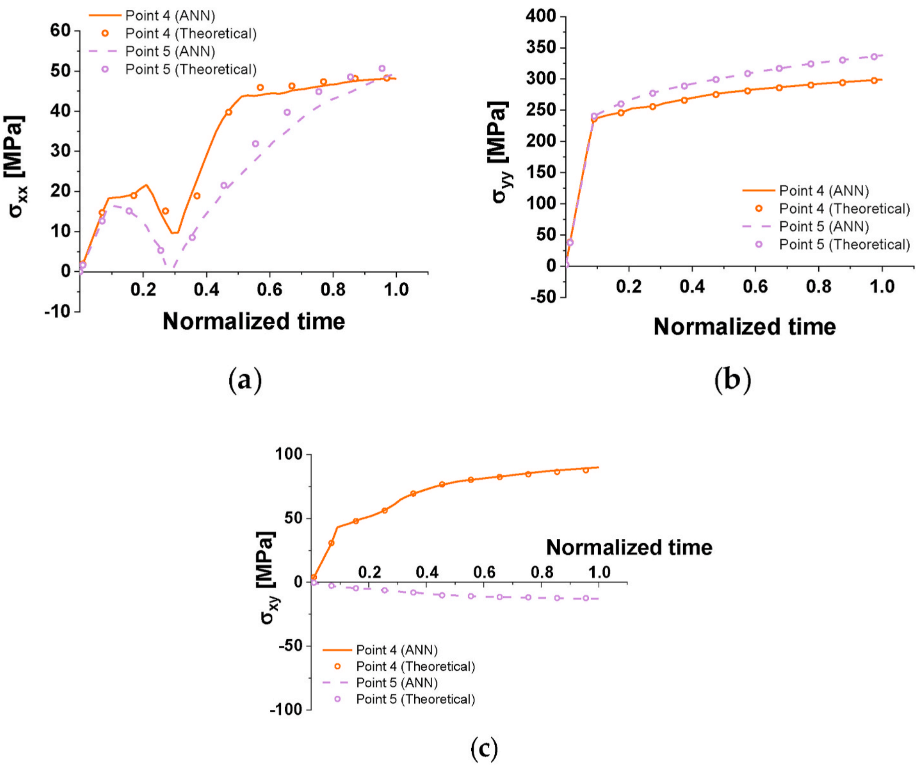

Fig. 15. Comparison of stress responses of two points in the fillet region predicted by the theoretical constitutive model and the ANN-based constitutive model: (**a**) normal stress along the x-direction; (**b**) normal stress along the y-direction; (c) shear stress.

noticeable difference. Comparing the CPU time, the theoretical model takes 478s and the ANN model takes 425s, which is 11% of time reduction. This result shows that the ANN model has computational efficiency in comparison with the conventional theoretical model. It is caused by the feed-forward nature of the ANN model. In other words, the ANN model does not require iterative computation in comparison to the theoretical model.

5. Conclusions

In this paper, an ANN-based constitutive model was developed to replace the nonlinear iterations for plastic correction schemes during the stress-integration method under von-Mises yield function and isotropic hardening law. A conventional physics-based model was kept utilized for elastic prediction, linear loading, and unloading. In this way, training can be done only for all the possible plastic loading paths, which makes the problems independent. The return mapping scheme combined with the proposed ANN-based constitutive model was used in the principal stress space. Training data were numerically generated by using the conventional constitutive equations. Training data represents not only a single problem but also the overall feature of the elastoplastic constitutive relation.

After training had been completed, the ANN-based constitutive model was implemented on a commercial FE code, ABAQUS2016/STANDARD. For verification purpose, single-element simulation under uniaxial tension, simple shear, and the plane strain were conducted. Even though the aforementioned four loading paths themselves were not trained directly, the ANN-based constitutive model predicts strain-stress response accurately, which implies the ANN-based constitutive model learned the elastoplastic constitutive relation. On the other hand, to verify the feasibility of multi-element simulation, a dog-bone tensile test simulation was conducted. It was verified that the ANN-based constitutive model describes both uniaxial tension in the gauge section and nonlinear loading paths in the fillet regions well. Finally, the ANN-based constitutive model was applied for circular cup drawing simulation. The predicted cup height profiles show a very small difference between the ANN-based constitutive model and the theoretical constitutive model. In terms of computational efficiency, the ANN-based constitutive model achieved 11% of time reduction in comparison with the theoretical constitutive model. This computational efficiency of the ANN-based constitutive model comes from the feed-forward nature of ANN. In contrast to the conventional theoretical constitutive model, the ANN model performs stress integration without any iterative computation.

For future work, an ANN-based constitutive model for advanced anisotropic yield function will be developed to capture anisotropic plastic behavior. For the application in FE analysis, drawing simulation and earing prediction will be conducted. In succession, the modeling accuracy and computational efficiency of the ANN-based constitutive model will be evaluated for comparison with the theoretical constitutive model.

CRediT authorship contribution statement

Dong Phill Jang: Writing - original draftWriting – original draft, contributed to implement the new model to FE method as well as

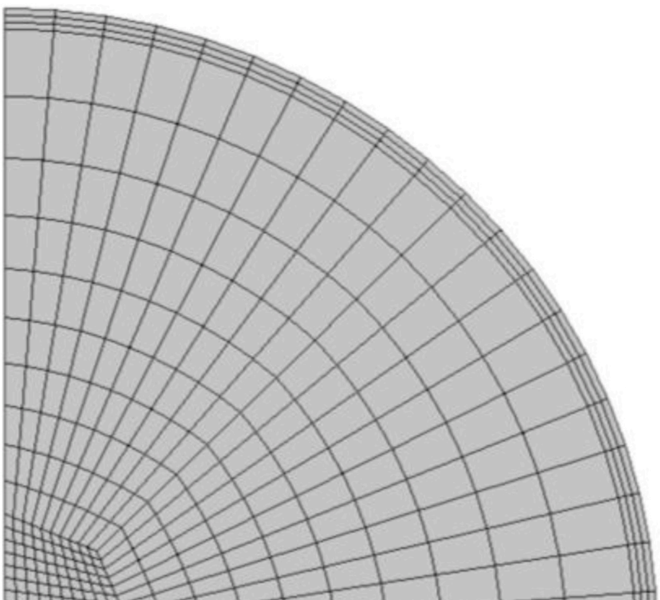

Fig. 16. Mesh design for the sheet blank.

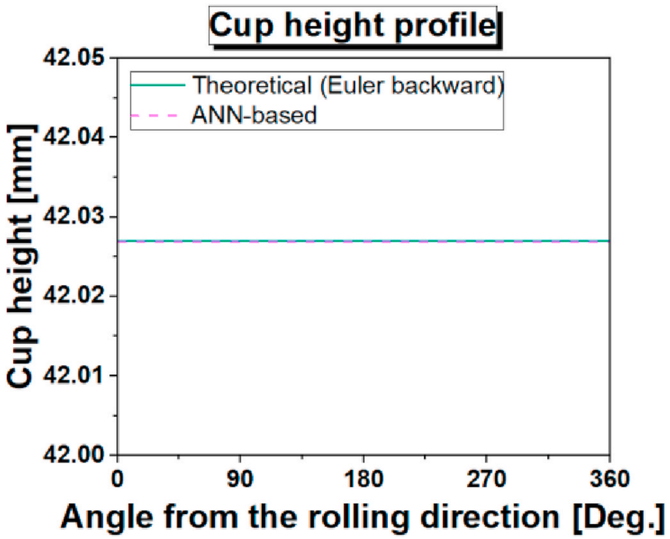

Fig. 17. Comparison of the cup height profiles predicted by the theoretical constitutive model and ANN-based constitutive model.

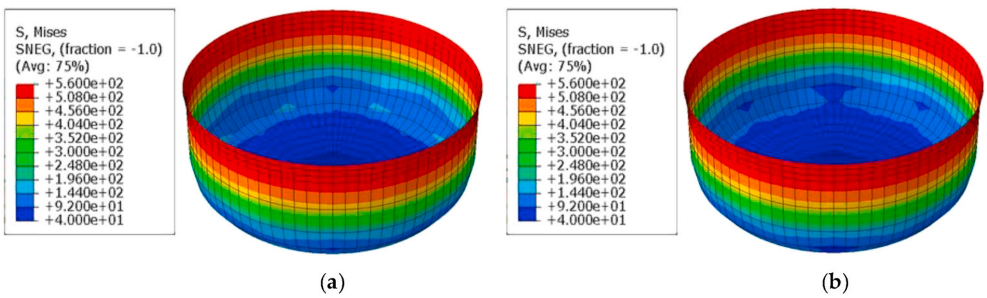

Fig. 18. Comparison of the effective stress contours predicted by: (**a**) the theoretical constitutive model; (**b**) ANN-based constitutive model.

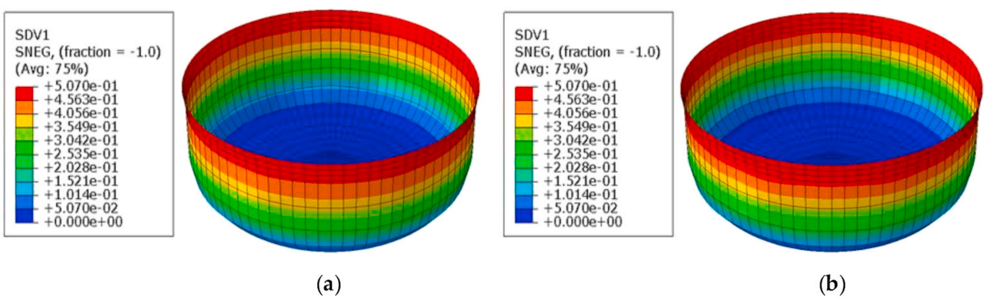

Fig. 19. Comparison of the effective plastic strain contours predicted by: (**a**) the theoretical constitutive model; (**b**) ANN-based constitutive model.

preparing for an initial draft. **Piemaan Fazily:** Writing - original draftWriting – original draft, contributed to improve the developed method with helping initial draft and revision. **Jeong Whan Yoon:** Supervision, Writing - original draftWriting – original draft, contributed to develop the new model with supervision and writing the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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